HIGH PERFORMANCE COMPUTING 101

HPC Storage @ IU (Module 3)

BRING ON TOMORROW

Module 3 — Lesson 2

Using IU HPC Files Systems

BRING ON TOMORROW

Overview

- Home directories
- HPC file systems at IU
- Pick the right system for your work
- Quota (the `quota` command)
- Scholarly Data Archive

Home directories

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Home Directories

- HPC setup always has server nodes that need to mount networked filesystems with identical pathing

- If the home directory is /home
  - the server exports whatever /home points to (e.g. /homedirs)
  - the client mounts it as /home

- Linux *symbolic links* are used to simplify/organize paths

```
[citrain@h2 ~]$ ls -ld /N/u/citrain
lrwxrwxrwx 1 root root 19 Dec 19 15:06 /N/u/citrain -> /N/home/c/i/citrain
```

- An example of mapping for organizational reasons on the storage server

Home Directories

- Look at your home directory on Quartz
  - Log in (or run `cd ~`)
  - Run the `pwd` command
    - You should see something like `/N/u/username/Quartz`
- Two important things to know:
  - All your home directories for HPC machines are in `/N/u/username/`, where username is your IU Account name
    - A subdirectory for each system you have accounts on will live there
      - e.g. `/N/u/username/Quartz` and `/N/u/username/BigRed200`

Home Directories

- Occasionally, you need to debug what physical path you're in
  - The *pwd* command will tell you the logical path if you got there via symlinks

  - Commands `pwd -L` & `pwd -P` give logical & physical working directories, respectively, e.g.

```
[citrain@h2 ~]$ pwd
/N/u/citrain/Quartz

[citrain@h2 ~]$ pwd -L
/N/u/citrain/Quartz

[citrain@h2 ~]$ pwd -P
/geode2/home/u020/citrain/Quartz
```

You can see that the logical path, mentioned above, is /N/u/username/Quartz.

The physical path, after resolving any symbolic links (two layers you can't see), is rooted in the Geode 2 mount point.

HPC file systems at IU

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HPC File Systems at IU

- Mount points to know for the HPC file systems
  - /N/slate — Slate personal HPC storage
  - /N/scratch — Slate scratch space for compute I/O
  - /N/project — Slate Project HPC storage (large data)
  - /geode2/projects — Geode Project HPC storage (large data)
  - /geode2/home — Geode home/login directory space

HPC File Systems at IU

- IU has three software stacks in production for file systems
  - IBM Storage Scale
    - still referred to as GPFS (General Parallel File System)
    - serves home directories and Geode Project
    - managed by the Research Storage group in RT
  - Lustre
    - Lustre is an open-source software stack w/ vendor support
    - serves Slate, Slate Scratch and Slate Project
    - managed by the High Performance File Systems group in RT
  - VAST
    - a new, high performance storage strategy being deployed by the Research Storage group

HPC File Systems

- Modern HPC file systems are parallel, cluster-based
    - storage is served from clusters of servers
    - typically each server manages arrays of storage drives
        - frequently storage servers are paired, to support failover
    - arrays of drives are usually managed via specialized controllers
        - controllers themselves are typically paired, for failover
- The storage system hierarchy will look much like a tree
    - storage drives will be the leaves on that tree
    - the drives are typically arranged as redundant RAID arrays

HPC File Systems

- RAID stands for "redundant array of inexpensive disks"

- We won't go into depth about RAID, but the idea is straightforward
    - if you one one drive, you get one-drive storage capacity
    - if you have N, you get N space, but no safety margin if one dies

- Imagine you have N drives, but you change how you use them
    - you can store N-1 drive capacity on N drives and lose one
    - you can store N-2 drive capacity on N drives and lose two

- So-called parity data can be used to recreate lost data

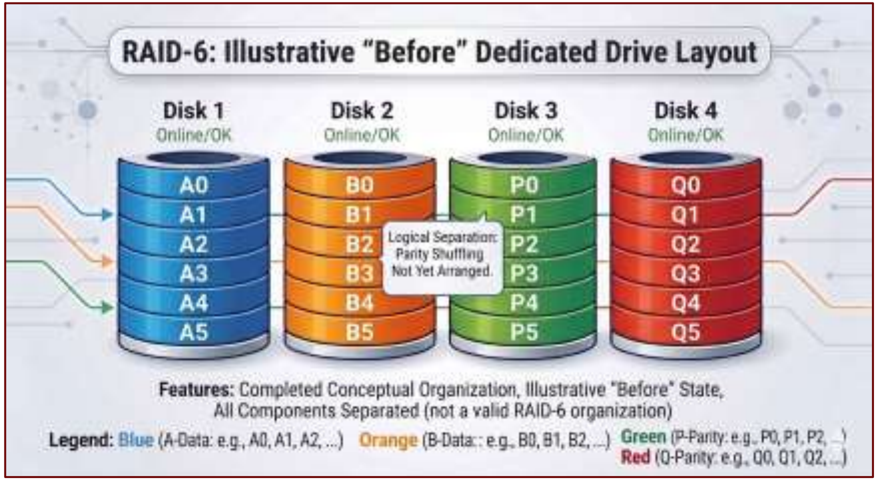
RAID Parity and Redundancy

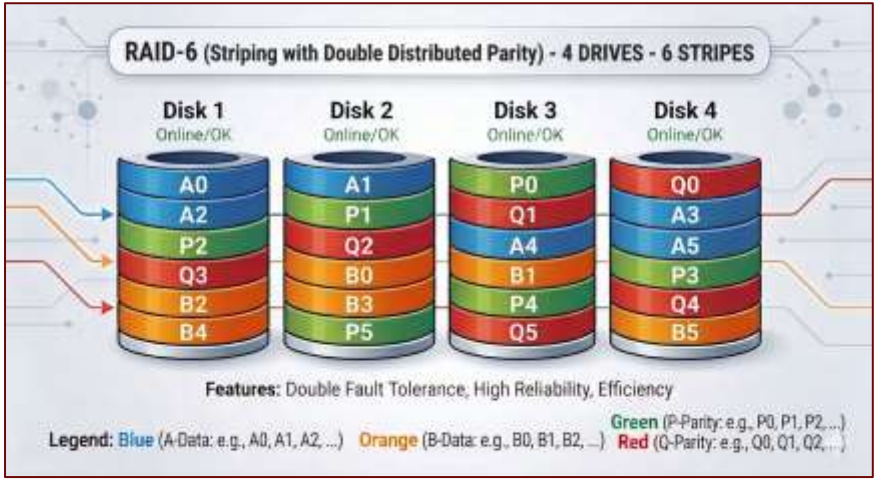
- For RAID-5, single-parity redundancy, recovery is possible from a single failed disk
  - RAID-6 double parity allows recovery of two disks

- The minimum RAID-6 config is 4 drives to get redundant storage of 2-drive capacity

- The process requires chunking the data and striping the data and corresponding parity across drives to allow for a failed drive
  - A and B in next image are purely for illustration of the logic
  - P and Q are the two parity values needed for RAID-6 recovery

RAID Parity and Redundancy





HPC File Systems

- RAID-6 is a so-called double-parity arrangement that allows two drives in a group to fail before data is lost
  - this is a typical HPC storage arrangement
  - additional "hot-spare" drives will automatically jump in on failure
  - spares are rebuilt from parity info on remaining drives, to replace failed drive

- Cluster file systems will use multiple RAID-6 arrays
  - **each array gets represented interally as a single virtual drive**

- This virtual drives are stitched together to form file systems

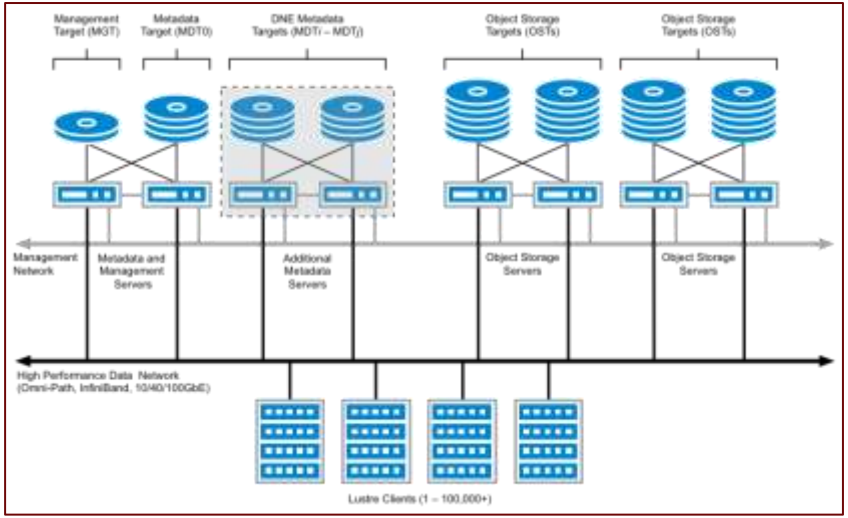
Key Components of a Parallel File System

- At the business end of the file system, there are drawers upon drawers of hard drives
  - Image to the right shows a typical storage drawer within the Slate systems
    - it holds 84 hard drives!
    - can support 24 TB drives, but probably has 16 TB drives

- Image at top is faceplate for a DDN SFA18kx drive controller
  - Can manage 22 84-drive drawers!
  - That's 1848 hard drives per controller!!
- Since controllers are paired for redundancy, normal behavior would max at 11 drawers

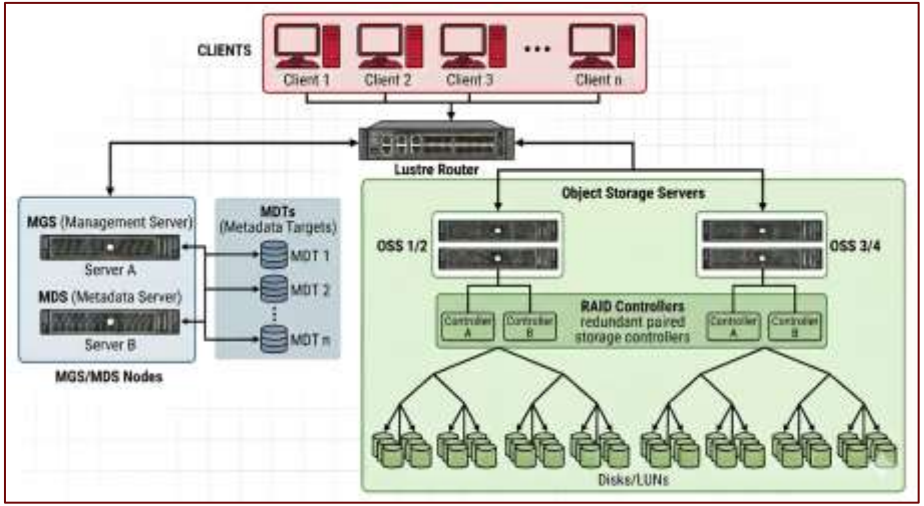
Key Components of a Parallel File System

- Components of parallel file systems are
  - Management server
  - Metadata server(s)
  - Storage server(s)
- Many storage systems are object based and provide a filesystem interface
- **On the client side, all this looks like one file system**
  - no different than a single virtual drive



Key Components of a Parallel File System

- Lustre file systems at IU use Infiniband between Lustre servers
- A Lustre router sits between clients that connect via 10/40/100 GbE
- Each Lustre file system is presented as a single path on client machines: e.g. /N/slate
  - Same for other HPC file systems



Pick the right system for your work

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Using IU HPC File Systems

- Pick the right system for your work
  - Criterion 1: "Compute against" Lustre file systems
    - Scratch, Slate and Slate Project are designed to handle the I/O directly from a large number of HPC jobs at one time
    - DO NOT read/write from/to your home directory with your SLURM jobs
      - your performance will be variable-to-substandard
      - it may bog down the experience of all users on the system(s)
    - Use local disk (/tmp) where possible

Using IU HPC File Systems

- Pick the right system for your work
  - Criterion 2: Scratch and local disk are ephemeral
    - Data on /tmp may not survive a reboot; assume it won't
    - Data on /tmp is local to the node it's written on
    - /N/scratch gets purged: **files not accessed in 30 days vanish**

    - If you need to store many small files (thousands), archive them to tar or zip files.
      - It is convenient and practical for keeping track of purge dates
    - Each file requires an inode and associated metadata
      - that adds up across millions of files and uses limited storage

Using IU HPC File Systems

- Pick the right system for your work
  - Criterion 3: Where to keep your project data longer term?
    - longer term here means more than 30 days
    - If you have less than 1600 GB, use /N/slate
    - For > 1.6 TB and less than 15 TB, apply for Slate Project space
      - see the KB for assistance; up to 15 TB is cost free
    - For more than 15 TB, consider both Slate Project & Geode Project; see the KB for both

Using IU HPC File Systems

- Use all of your storage options
  - just use them wisely

  - note that none of the HPC file systems are backed up
    - **redundant doesn't mean backed up; if you delete it, it's gone**

- Imagine that you work with a research group that has a Slate Project storage area called /N/project/super-research-1

Using IU HPC File Systems

- Use your home directory and your Slate directory for personal work that others don't need access to

- Use Scratch and local disks for your runtime job I/O, as appropriate for the work you're doing

- Use the Slate Project account to read/write/share data and work with others in your research project

  - keep in mind that others in the project will have the ability to overwrite/delete data you write there, as it's shared

UNFINISHED SLIDE (DELETE THIS MARKER WHEN FINISHED)

VAST Placeholder

Talk to Charlie?

What's happening with VAST

- Just home directories, or geode project as well?
- Can we compute against VAST?

Quota (the `quota` command)

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Quota

- We've covered limits on storage space

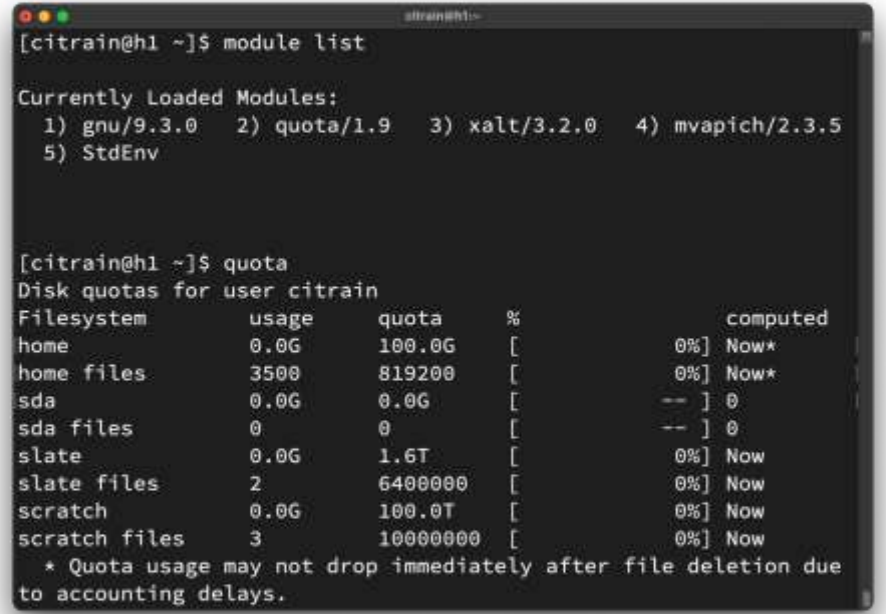
- How do you know what the limits are on your account and how much you have used?

    - The limits are referred to as "quotas"
    - There is a quota command that will give you this information

    - IU HPC accounts use a customized quota command
        - loaded by default, as a module (called "quota")

Quota

- The `module list` command shows quota as a loaded module
- Running `quota` shows user quotas on file systems the user has access to
  - Note there are inode quotas
  - inode limits are max # files
- sda here means Scholarly Data Archive (more on that next)



Scholarly Data Archive

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Scholarly Data Archive

- The Scholarly Data Archive (SDA) is a large tape and disk data archiving system based on the HPSS software stack (see KB)
  - Approximately 79 petabytes (PB) in tape storage
  - Geographically distributed between Bloomington and Indy
  - Default is dual copies of data on each tape system between sites

- SDA does not present a file system interface
  - data can be moved back and forth via
    - hsi/htar command line client)
    - SFTP (command line or GUI)
    - Globus file transfer

Reflection

- You now know:
  - how IU organizes home directories and what `/N/u/username/Quartz` means
  - the mount points, software stacks, and RAID architecture of IU's HPC file systems
  - how to choose the right file system for each part of your workflow
  - how to read your storage usage and limits with the `quota` command
  - what the Scholarly Data Archive is and when to use it
- Next module covers Moving Data (Module 4)
- Be sure to complete the reflection for this lesson